



Monitoring-based identification of nonlinear aerodynamic damping in multi-mode vortex-induced vibrations of a stay cable

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ABSTRACT

New issues of wind-induced cable vibration, i.e., multi-mode vibrations, increasingly occur in long-span bridges, which are closely associated with underlying damping features. In this study, an analytical method is proposed to identify nonlinear aerodynamic damping in multi-mode wind-induced cable vibrations in fields, involving the assessment of total and mechanical damping. The method is available to extract a series of mono-component resonances/decays from the measured signals based on the analytical mode decomposition (AMD) and random decrement technique (RDT), and then to determine the damping parameters using the Hilbert transform (HT) combined with the smoothing spline or linear least-square fitting. Furthermore, this method is applied to address the vortex-induced vibrations (VIV) of an ultra-long stay cable on the Sutong Bridge. Firstly, the vibrations are reported in terms of a typical event and one-year statistics. Subsequently, the damping parameters are identified to illustrate the in-situ aeroelasticity. The results show that the cable exhibits high-frequency vibrations primarily in the 41st to 43rd modes, and the aerodynamic damping exhibits significant amplitude-dependent nonlinearity. Finally, the damping identifications are employed to reveal the onset mechanism for the vibrations, and a Scruton number (Sc) criterion, i.e., $Sc > 5.4$, is suggested for mitigating the vibrations.

1. Introduction

Bridge stay cables are potentially excited into dramatic vibration of various types, including vortex-induced vibration (VIV) (Argentini et al., 2016; Urushadze and Pirner, 2018; He et al., 2022), rain-wind-induced vibration (RWIV) (Daniotti et al., 2021), and galloping (An et al., 2021). These vibrations, for shorter stay cables (e.g., span $L < 200\text{m}$), are typically dominated by a single mode of the first few during a specific vibration event. As stay cables continue to grow in length, however, high-order multi-mode vibrations are increasingly observed in recent years. These vibrations are not effectively controlled with the traditional countermeasures, resulting in more undue stress and thereby more severe fatigue damage. This emerging issue poses significant challenges to engineering designers.

Field monitoring is a powerful approach to investigating the multi-mode wind-induced vibrations of stay cables. The multi-mode VIVs on a stay cable ($L = 323\text{m}$) were observed by Chen et al. (2019), indicating that the vibrations primarily exhibited over the 18th to 40th modes under wind velocities ranging from 4 to 6 m/s. The vibration

characteristics of a stay cable ($L = 576.77\text{m}$) were identified by Ge and Chen (2019), finding that the cable experienced RWIVs in the 13th mode and VIVs in the 43rd and 44th modes. The multi-mode VIVs of a stay cable ($L = 349.2\text{m}$) were reported by Kim et al. (2022), revealing the vibrations within the frequency ranging from 10 to 40 Hz in a wide range of wind velocities. The multi-mode vibrations of a stay cable ($L = 512\text{m}$) were presented by Wang et al. (2022a), summarizing that the VIVs and RWIVs occurred over frequency bands of 2.4–12.3 Hz and 2.2–4.1 Hz, respectively. The dynamic responses of stay cables ($L = 337.4\text{m}$, 541.1m) under extreme winds were analyzed by H. Zhang et al. (2022), showing the multi-mode vibrations with frequencies above 3 Hz. To sum up, these reports have revealed the vibration characteristics of longer cables, whereas the underlying aeroelastic mechanism remains unclear.

Multi-mode cable vibrations in fields can be addressed well using aeroelastic dynamics modeling, in which, aerodynamic damping, governing the energy transfer from air to specific cable modes, is of significance for predicting and mitigating cable vibrations. Great efforts have been experimentally/numerically made to identify aeroelastic

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parameters of sectional circular cylinders in terms of VIV (Lupi et al., 2018; R. Zhang et al., 2023), RWIV (Jing et al., 2018) and dry-cable galloping (Jafari and Sarkar, 2019). With these identifications, Jafari and Sarkar (2019) estimated the minimum damping required to prevent the dry-cable galloping up to the designed wind velocity; R. Zhang et al. (2023) predicted the VIV amplitudes of a circular cylinder under various mass-damping conditions. Nevertheless, the identifications are not directly applicable to in-situ cables due to involving multi-mode couplings, large aspect ratios, non-uniform turbulent winds, etc. In this sense, some attempts have been made to identify aerodynamic damping of stay cables based on field monitoring. Ni et al. (2007) estimated the segment-averaged aerodynamic damping of the first three modes of a stay cable during RWIVs. Hua and Zuo (2019) evaluated the van der Pol-type nonlinear aerodynamic damping of the second to fourth modes of stay cables under RWIVs. However, there are currently rare field reports on nonlinear aerodynamic damping identifications of stay cables, especially for high-order multi-mode vibrations.

An alternative scheme for identifying aerodynamic damping is to first derive the total damping by measuring the growing or decaying rate of the vibration evolution and then subtract the corresponding mechanical damping (Mashnad and Jones, 2014). However, two aspects of issues are encountered when identifying multi-mode aerodynamic damping in fields, namely the multi-mode coupling in the vibration signals and the unknown mechanical damping of in-situ cables. To address these issues, some advanced techniques in signal processing and system identification can be considered. For instance, the techniques, e. g., the wavelet transform, empirical mode decomposition (EMD) (Huang et al., 1998), and analytical mode decomposition (AMD) (Chen and Wang, 2012), are available for decomposing multi-frequency components; the methods, e. g., the random decrement technique (RDT) (Ibrahim, 1977) and natural excitation technique (NExT) (James et al., 1995), are feasible for modal identification. These techniques can be integrated to identify the multi-mode linear mechanical systems under operational conditions (He et al., 2011; Lardies and Ta, 2011; Wen et al., 2018). However, there is still a lack of attempts to identify multimode aeroelastic systems of nonlinearity, especially for closely-spaced modal frequency systems. Consequently, it is crucial to establish an analytical framework well-suited for identifying nonlinear aerodynamic damping in multi-mode wind-induced cable vibrations in field environments.

This study presents a monitoring-based investigation of the multi-mode VIVs of an ultra-long stay cable ($L = 551.01\text{m}$) by applying a proposed method for aerodynamic damping identification. The remainder of this paper is organized as follows. Section 2 introduces the details of the analytical method. Section 3 presents a case study on the Sutong Bridge. Finally, Section 4 summarizes the paper.

2. Methodology

Based on the dynamics of flexible cables and the modeling of wind loads, an analytical method is proposed to identify the multi-mode nonlinear aerodynamic damping of stay cables in the context of field monitoring, which is achieved by identifying the total and mechanical damping.

2.1. Theoretical background

For an inclined stay cable subjected to dynamic wind loads, the equations of motion, taking the in-plane vibration as an example, can be obtained by employing the Hamilton principle (Wang et al., 2005):

$$m \frac{\partial^2 y}{\partial t^2} + c \frac{\partial y}{\partial t} - T \frac{\partial^2 y}{\partial x^2} - \frac{\lambda^2 T}{L^3} \int_0^L y dx = F(x, t) \quad (1)$$

where $y(x, t)$ is the dynamic displacement; m and c are the mass and damping coefficient per unit length of the stay cable, respectively; T is the initial tension; L is the cable span; $F(x, t)$ is the external wind loads

on the stay cable; λ^2 is the dimensionless sag parameter, defined as follows:

$$\lambda^2 = \frac{(mgL \cos \alpha)^2 EA}{T^3 L_e / L} \quad (2)$$

where g is the gravity acceleration; α is the inclined angle; E is Yong's modulus; A is the cross-section area; $L_e = L[1 + (mgL \cos \alpha / T)^2 / 8]$ is the effective length of the stay cable in the static configuration. By neglecting the coupling between in-plane and out-of-plane motions, the wind loads on the oscillating cable at lock-in, i.e., the nonlinear self-excited aerodynamic forces, can be written as (Simiu and Scanlan, 1996; Jafari et al., 2020):

$$F(x, t) = \frac{1}{2} \rho U^2 (2D) \left\{ Y_1(K) \left[1 - \varepsilon(K) \frac{y^2}{D^2} \right] \frac{\dot{y}}{U} + Y_2(K) \frac{y}{D} \right\} \quad (3)$$

where ρ is air density; U is the wind speed; D is the diameter of the stay cable; $K = \omega D / U$ is the reduced frequency with ω being the vibration frequency; Y_1^* and ε are the parameters associated with linear and nonlinear aerodynamic damping terms, respectively; Y_2^* is the parameter related to aerodynamic stiffness term. These parameters are functions of the reduced frequency and should depend on the yaw angle and turbulences of incoming winds in the fields. Strictly speaking, the self-excited force on the cable derives from not only in-plane but also out-of-plane motions, and the out-of-plane motion plays an important role in the galloping of cables (Jafari and Sarkar, 2020). Whereas, in Eq. (3), the out-of-plane motion's effects are neglected for convenience due to its relatively small participation in the VIVs of more concern herein.

According to the Galerkin method, the dynamic response $y(x, t)$ can be expressed in terms of a finite series (Wang and Zhao, 2009):

$$y(x, t) = \sum_{i=1}^n \varphi_i(x) q_i(t) \quad (4)$$

where $q_i(t)$ is the generalized coordinate and $\varphi_i(x)$ is the shape of the i th mode; n is the number of modes of concern. Substituting Eq. (4) into Eq. (1) and dropping the negligible aerodynamic stiffness term, one yields the following matrix equation of motion:

$$\mathbf{M} \ddot{\mathbf{q}} + (\mathbf{C}_s + \mathbf{C}_a) \dot{\mathbf{q}} + \mathbf{K}_s \mathbf{q} = \mathbf{0} \quad (5)$$

In Eq. (5), $\mathbf{M} = \text{diag}[m_i]$, $\mathbf{C}_s = \text{diag}[2m_i \xi_{si} \omega_{si}]$ and $\mathbf{K}_s = \text{diag}[m_i \omega_{si}^2]$ are the generalized mass, damping and stiffness matrices of the stay cable, respectively, where $m_i = m \int_0^L \varphi_i^2 dx$ is the i th modal mass; ξ_{si} and ω_{si} are the i th modal mechanical damping ratio and natural frequency, respectively; $\mathbf{C}_a = \text{diag}[2m_i \xi_{ai} \omega_{si}]$ is the generalized aerodynamic damping matrix, with ξ_{ai} being the i th modal aerodynamic damping ratio, which is defined as:

$$\xi_{ai} = -\frac{1}{2\mu K_{ref}} Y_{1,g} (1 - \varepsilon_g \eta_i^2) \quad (6)$$

where $\mu = m / (\rho D^2)$ is the mass ratio of the stay cable to the placed air; $K_{ref} = \omega_{si} D / U_{ref}$ is the reduced frequency for reference with U_{ref} being a referenced wind velocity; $Y_{1,g}$ and ε_g are the generalized linear and nonlinear aerodynamic damping parameters, respectively, which are expressed as follows:

$$Y_{1,g} = \frac{\int_0^L \frac{U}{U_{ref}} Y_1 \varphi_i^2 dx}{\int_0^L \varphi_i^2 dx}, \quad \varepsilon_g = \frac{\int_0^L \frac{U}{U_{ref}} Y_1 \varepsilon \varphi_i^4 dx}{Y_{1,g} \int_0^L \varphi_i^2 dx} \quad (7)$$

and $\eta_i = q_i / D$ is the normalized generalized coordinate of the i th mode. Moreover, according to the harmonic balance method (Chen, 2013), the modal displacement-dependent aerodynamic damping in Eq. (6) can be equivalently transformed into the functions of modal displacement amplitude for convenience of parameter identification, as follows:

$$\xi_{ai} = -\frac{1}{2\mu K_{ref}} Y_{1,g} \left(1 - \frac{1}{4} \varepsilon_g A_{\eta_i}^2 \right) \quad (8)$$

where A_{η_i} is the i th normalized modal displacement amplitude.

2.2. Analytical approach

For a given stay cable, the mass and stiffness matrices in Eq. (5) can be determined through the design data, whereas the damping parameters are unknown and require identification using particular approaches.

2.2.1. Signal decomposition

Stay cable vibrations involve multi-mode participation as well as quasi-static responses. In light of this, the AMD proposed by Chen and Wang (2012) is employed to extract a series of sub-signals of individual modes. According to the theorem, a real signal $y(t)$ consisting of n significant frequency components ($\omega_1, \omega_2, \dots, \omega_n$) can be decomposed into a series of time series whose Fourier spectra over mutually exclusive frequency ranges:

$$y(t) = \sum_{i=1}^n y_i(t) \quad (9)$$

where each time series $y_i(t)$ has a narrow frequency band and can be defined by:

$$y_i(t) = s_1(t), \dots, y_i(t) = s_i(t) - s_{i-1}(t), \dots, y_n(t) = y(t) - s_{n-1}(t) \quad (10)$$

In Eq. (10), the new series $s_i(t)$ could be determined as follows:

$$s_i(t) = \sin(\omega_{bi}t) H[y(t) \cos(\omega_{bi}t)] - \cos(\omega_{bi}t) H[y(t) \sin(\omega_{bi}t)] \quad (11)$$

where $H[\cdot]$ denotes the Hilbert transform operator, and ω_{bi} is the bisecting frequency between ω_i and ω_{i+1} . The selected frequency ranges should capture the target spectral bands, as illustrated in Fig. 1. For stay cable systems, the natural frequency approximately increases by multiple integers of the fundamental frequency, and hence the bisecting frequencies can be initially defined as the natural frequencies minus or plus half of the frequency interval, i.e., $\omega_{si} \pm \Delta\omega_s/2$. However, the spectral peaks of the measured responses may deviate slightly from the calculated natural frequencies. In such cases, the bisecting frequencies need to be redefined accordingly.

Moreover, the decomposed modal components $y_i(t)$ at the measured point $x = x_0$ can be scaled to yield the corresponding generalized modal responses according to the mode shapes:

$$q_i(t) = y_i(t) / \varphi_i(x_0) \quad (12)$$

In Eq. (12), the mode shape of a stay cable can be approximately expressed as: $\varphi_i(x_0) = \sin(i\pi x_0 / L)$.

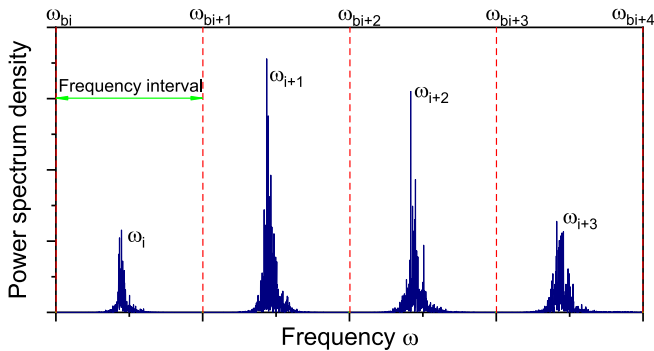


Fig. 1. Selection of bisecting frequencies in the analytical mode decomposition for stay cable systems.

2.2.2. Transient response

Two types of transient responses, i.e., wind-induced resonance and free-to-decay signature, are the basis for identifying total and mechanical damping, respectively. The evolution of the wind-induced resonance generally includes three stages, namely growth-to-resonance, fully-developed, and resonance-to-decay stages, as depicted in Fig. 2. This evolution could reflect the energy-trapping/dissipating properties of stay cables from/to their surrounding winds through the cable-wind interaction.

On the other hand, the RDT (Ibrahim, 1977) is adopted to determine the free-to-decay signatures from the ambient-excited cable responses in the time domain. The theoretical basis for this approach is described as follows. According to Duhamel's integral principle, the dynamic response $y_i(t)$ at a location ($x = x_0$) for the i th mode of a linear system subjected to arbitrary loads consists of free decay and forced vibration, as follows:

$$y_i(t) = e^{-\xi_{si}\omega_{di}t} \left\{ y_i(0) \left[\cos(\omega_{di}t) + \frac{\xi_{si}\omega_{si}}{\omega_{di}} \sin(\omega_{di}t) \right] + \frac{\dot{y}_i(0)}{\omega_{di}} \sin(\omega_{di}t) \right\} + \varphi_i(x_0) \int_0^t h_i(t-\tau) F_i(\tau) d\tau \quad (13)$$

where $y_i(0)$ and $\dot{y}_i(0)$ are the initial displacement and velocity at $t = 0$, respectively; $\omega_{di} = \omega_{si} \sqrt{1 - \xi_{si}^2}$ is the i th modal damped frequency; F_i is the generalized force, and $h_i(t - \tau)$ is the impulse response function of the i th mode. Supposing that the external force is a zero-mean stationary Gaussian process, the forcing convolution integral in Eq. (13) over time is thus expected to be zero. In this sense, the transient decay can be determined by averaging the ensemble data series under the specific initial condition ($y_i = a, \dot{y}_i = b$), given as follows:

$$E[y_i(t + \tau) | y_i(t) = a, \dot{y}_i(t) = b] = A_{y,i} e^{-\xi_{si}\omega_{di}\tau} \sin(\omega_{di}\tau + \varphi_i) \quad (14)$$

In Eq. (14), $A_{y,i}$ and φ_i are the initial amplitude and phase, respectively. Considering the limited duration of the signal, the random decrement signature $\delta_i(\tau)$ for the i th mode can be extracted in a new time coordinate system by averaging the selected N signal segments:

$$\delta_i(\tau) = \frac{1}{N} \sum_{k=1}^N y_i(t_k + \tau) \quad (15)$$

In practice, the initial conditions are typically given as the specific displacement or acceleration, and thus the free vibration due to the initial velocity will be averaged out. The process of RDT is illustrated in Fig. 3.

2.2.3. Parameter identification

Once the single-mode transient resonance/decay responses are extracted, the Hilbert transform (HT) is applied to identify the modal parameters of either linear or nonlinear systems. The HT of a time series $y_i(t)$ is defined as follows:

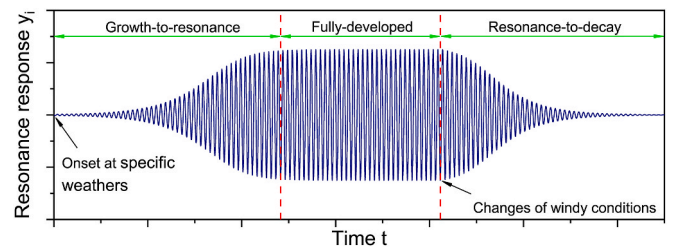


Fig. 2. Schematic illustration of wind-induced resonance evolution during a typical vibration event of a stay cable.

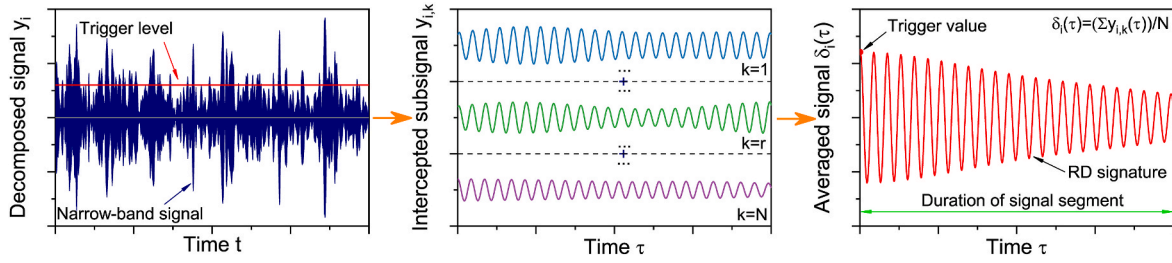


Fig. 3. Extraction of the free-to-decay response (random decrement signature) from the narrow-band ambient-excited signal of stay cables by the random decrement technique.

$$\tilde{y}_i(t) = H[y_i(t)] = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y_i(\tau)}{t - \tau} d\tau \quad (16)$$

Then, an analytic signal associated with $y_i(t)$ can be determined as follows:

$$Y_i(t) = y_i(t) + i\tilde{y}_i(t) = A(t)e^{i\theta(t)} \quad (17)$$

where $A(t)$ and $\theta(t)$ are the instantaneous amplitude (i.e., the envelope) and phase angle of $y_i(t)$, respectively, and they are defined as:

$$A(t) = [y_i^2(t) + \tilde{y}_i^2(t)]^{1/2}, \theta(t) = \tan^{-1}[\tilde{y}_i(t) / y_i(t)] \quad (18)$$

By applying the logarithmic and differential operators to Eq. (18), the instantaneous frequency and damping ratio can be determined from the following equations:

$$\omega(t) = \frac{d\theta_{fit}(t)}{dt}, \xi(t) = \frac{1}{\omega(t)} \frac{d \ln A_{fit}(t)}{dt} \quad (19)$$

where $\theta_{fit}(t)$ and $A_{fit}(t)$ are the instantaneous amplitude and phase after fitting, respectively. This fitting process aims to eliminate their local fluctuations due to the possible mixing among components of adjacent modes and the end effects of the HT. Specifically, smoothing spline fitting is employed to remove the spikes of the envelopes when identifying the total damping during wind-induced resonance, whereas linear least-square fitting is applied to the functions of $\theta(t)$ and $\ln A(t)$ when determining the natural frequency and mechanical damping ratio. Fig. 4 illustrates the two types of fitting in parameter identification.

The increasing/decreasing rate of wind-induced cable resonance is characterized as total damping, which is the sum of mechanical damping and aerodynamic damping (Main and Jones, 2001; Ni et al., 2007), and thus the aerodynamic damping is estimated as:

$$\xi_{ai} = \xi_{total,i} - \xi_{si} \quad (20)$$

where $\xi_{total,i}$ is the total damping ratio of the i th mode. Conceivably, wind-induced cable instability will occur if the total damping becomes zero or less than zero which is possible when the aerodynamic damping becomes negative with a magnitude equal to or exceeding that of the

mechanical damping.

2.3. Processing procedure

By applying the above analytical approach, the multi-mode nonlinear damping parameters can be identified based on the monitoring data, which mainly includes the following steps.

- (1) *Decomposing vibration signals*: The wind-induced resonances and/or ambient-excited responses are first selected from the monitoring data by detecting the response magnitude and wind conditions. Then, the Fourier spectra are employed to determine the modes of concern, along with the corresponding bisecting frequencies ω_{bi} . On this basis, the narrow-band signal can be extracted by performing the AMD on the concerned frequency ranges (see Eqs. (10) and (11)).
- (2) *Extracting transient responses*: For the wind-induced resonances, the decomposed signals are directly taken as the transient responses available for the following analysis. Whereas, for the ambient-excited responses, the RDT is still required to extract the free-to-decay signatures. To achieve this, (i) the trigger condition and the segment length are first set; (ii) the segmented signals are then gathered when the trigger condition is met; (iii) the free-to-decay signatures are obtained by averaging the segments (see Eq. (15)).
- (3) *Identifying damping parameters*: Based on the obtained two types of transient responses, the HT is first applied to determine their envelopes and instantaneous phases via Eq. (18). Then, the natural frequencies and mechanical damping ratios, as well as the time-varying total damping ratios, can be identified according to Eq. (19). Finally, the aerodynamic damping can be determined by subtracting the mechanical damping from the total damping (see Eq. (20)). The generalized nonlinear aerodynamic parameters are further obtained by fitting the amplitude-dependent aerodynamic damping (see Eq. (8)).

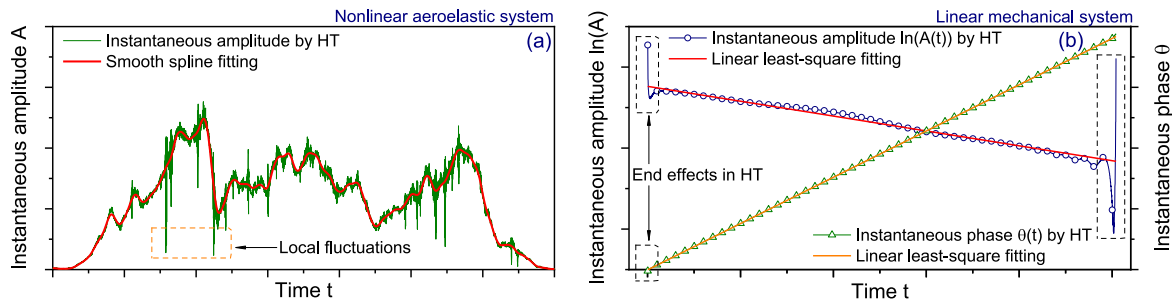


Fig. 4. Fitting of instantaneous amplitude and phase of the transient responses for parameter identification: (a) smoothing spline fitting; (b) linear least-square fitting.

3. Application to the Sutong Bridge

Based on the monitoring data of the Sutong Yangtze River Bridge in Jiangsu Province, China, the proposed method is applied to deal with the multi-mode vortex-induced vibrations of an ultra-long stay cable.

3.1. Description of the bridge

The Sutong Bridge is a cable-stayed bridge with a main span of 1088 m. The bridge deck is composed of a steel closed-box girder with a width of 41.0 m and a height of 4.0 m, which is supported by two inverted Y-shaped towers through a total of 272 stay cables arranged in two planes. The Sutong Bridge has been instrumented with a structural health monitoring (SHM) system after completion to evaluate the bridge's performances, as shown in Fig. 5. This monitoring system consists of 15 types of sensors with a total number of over one thousand. Particularly, four three-dimensional (3D) ultrasonic anemometers are installed at the bridge deck and the tower tops to record the wind velocity and direction. Furthermore, biaxial accelerometers are mounted at the bridge deck and the specific stay cables (depicted in Fig. 5) to measure their accelerations. In recent years, some stay cables of the bridge have experienced dramatic vibrations. To detect these vibrations, supplemental monitoring instruments are installed on this bridge (Yang et al., 2021; Wang et al., 2022a). Specifically, the accelerometers (DH1A202E, Donghua Tech.) are installed on the eight stay cables (denoted in Fig. 5) to record the in-plane and out-of-plane vibrations of the stay cables simultaneously. Also, it is noted that the anemometers are capable of measuring the wind velocity up to 45.0 m/s and wind direction ranging from 0° to 360° with a sampling frequency of 1.0 Hz, while the accelerometers have a wide measurement range of $\pm 4g$ and a much higher sampling frequency of 100.0 Hz, enabling the capture of possible high-frequency vibrations with greater precision.

This investigation focuses on an ultra-long stay cable of the Sutong Bridge, NJ32U, with a length of $L = 551.01$ m and a diameter of $D = 0.152$ m (aspect ratio $L/D = 3625$), as illustrated in Fig. 6. The stay cable is fabricated using parallel wires wrapped in polyethylene protective tubes covered with dimpled surfaces. More importantly, an MR damper is equipped in the plane of the stay cable to mitigate vibrations, and this damper has been thoroughly examined and repaired before the present monitoring. Also, high-damping rubber is attached to the cable anchorages at the bridge deck. An auxiliary steel wire rope is incorporated into the stay cable near the MR damper to partly participate the vibration control. More parameters of the stay cable are illustrated in Table 1.

3.2. Overview of cable vibrations

In 2019, the SHM system recorded over fifty high-frequency vibration events of the stay cable NJ32U, most of which occurred between May and September with the most endured and intense record taking place on September 13.

Fig. 7 shows the whole evolution of the typical vibration event of the stay cable observed on September 13, 2019. During this event, the mean wind velocities and directions are 7.1 m/s and 72.4° (yaw angle is $\beta = 28.2^\circ$) at the deck level and 10.7 m/s and 84.8° (yaw angle is $\beta = 15.8^\circ$)

at the tower top, respectively (see Fig. 7a and b). The recorded winds are remarkably steady without significant variation in wind directions, and the mean turbulence intensity at the deck level and the tower top are only 4.5 and 2.9 percent, respectively. Fig. 7c and d presents the corresponding time histories of the in-plane and out-of-plane accelerations of the stay cable. The in-plane vibration is triggered at about 19:40 and gradually increases to its maximum amplitude of 36 m/s^2 . Afterward, the vibration experiences rising and descending for about 2 h before eventually decaying out. Interestingly, the decay rate of the vibration is much larger than the growth rate. Specifically, it takes about 53 min from the onset to the maximum response, while the vibration completely decays in 30 min. On the other hand, the out-of-plane vibration develops similarly but with much smaller amplitudes. Furthermore, the evolutionary spectra of the stay cable accelerations are determined through the short-time Fourier transform (STFT), as illustrated in Fig. 7e and f. Both the in-plane and out-of-plane vibrations are characterized by three prominent frequencies around 10.4 Hz, corresponding to the 41st to 43rd modes of the stay cable, indicating obvious high-order multi-mode behaviors.

Subsequently, it is to elucidate the vibration mechanism of the stay cable based on the observations as follows. Due to the high-frequency features and dry conditions, the vibration is thus inferred to be excited by the Karman vortices in the cable wake. In this sense, the vortex-shedding frequency f_{vs} can be determined based on the Strouhal law:

$$f_{vs} = \frac{StU \cos \beta^*}{D} \quad (21)$$

In Eq. (21), St is the Strouhal number and is considered as 0.19 for the circular cylinder in the higher subcritical regime (R. Zhang et al., 2023); $\beta^* = \sin^{-1}(\sin \beta \cos \alpha)$ is the equivalent wind yaw angle (Jafari and Sarkar, 2019). Then, the time-averaged shedding frequencies f_{vs} at the deck level and tower top are 7.97 Hz and 12.94 Hz, respectively, and their mean (10.45 Hz) matches closely the observed dominant frequency of the stay cable vibration (10.4 Hz). Therefore, this event is attributed to VIV. To further prove this inference, Fig. 8 shows the variation of the vibration responses against the wind velocity during this event. The responses are observed so sensitive to incoming wind velocity that even a slight change can result in the onset or decay of the vibration. Furthermore, the dominant frequencies, corresponding to the maximum in the power spectrum density (PSD) function of the acceleration, are somewhat scattered at relatively low wind velocities but stabilize at about 10 Hz when the wind velocity exceeds 10 m/s, demonstrating the vortex lock-in behavior.

After analyzing the above vibration event, it is to make some statistics between the stay cable vibrations and windy conditions over the one-year monitoring data in 2019. The wind data from the tower-top anemometer is utilized as a reference due to its smaller disturbances. Moreover, the two turbulence parameters, i.e., turbulence intensity I_u and integral length L_u , are determined via the following equations (Simiu and Scanlan, 1996):

$$I_u = \frac{\sigma_u}{U}, L_u = \frac{U}{\sigma_u^2} \int_0^\infty R_u(\tau) d\tau \quad (22)$$

where σ_u is the standard deviation of the wind velocity; U is the mean

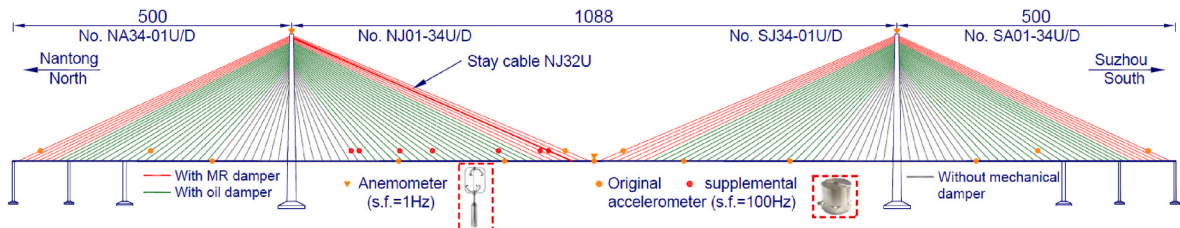


Fig. 5. Structural health monitoring system of the Sutong Yangtze River Bridge and partial instrumented sensors (unit: m).

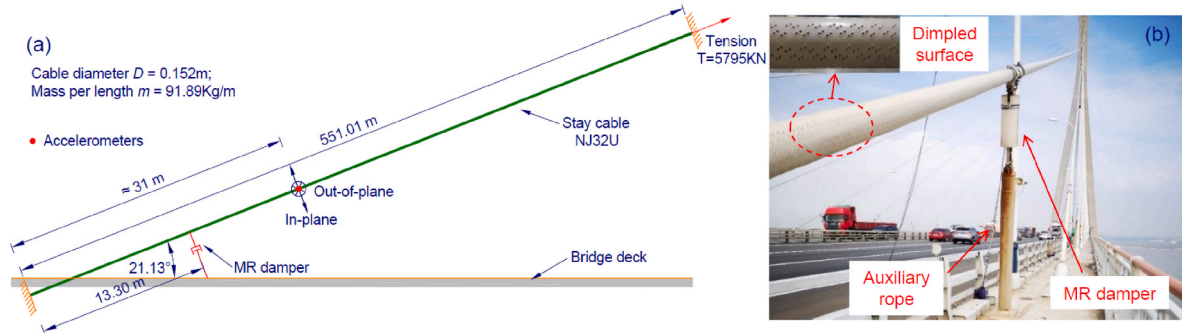


Fig. 6. Stay cable NJ32 of the Sutong Yangtze River Bridge: (a) technical illustration; (b) onsite photo.

Table 1

Parameters of the stay cable NJ32U of the Sutong Yangtze River Bridge.

L (m)	D (m)	α (°)	m (kg/m)	T (kN)	E (MPa)	A (m ²)	λ^2	$f_{s,1}^{in}$ (Hz)	$f_{s,1}^{out}$ (Hz)
551.01	0.152	21.13	91.89	5795	1.95×10^5	0.01022	2.20	0.259	0.241

Note that $f_{s,1}^{in}$ and $f_{s,1}^{out}$ are the in-plane and out-of-plane fundamental frequencies of the stay cable, respectively, and the measured in-plane fundamental frequency $f_{s,1}^{in}$ (0.259 Hz) is slightly larger than the frequency interval between modes (0.248 Hz) due to the cable sag.

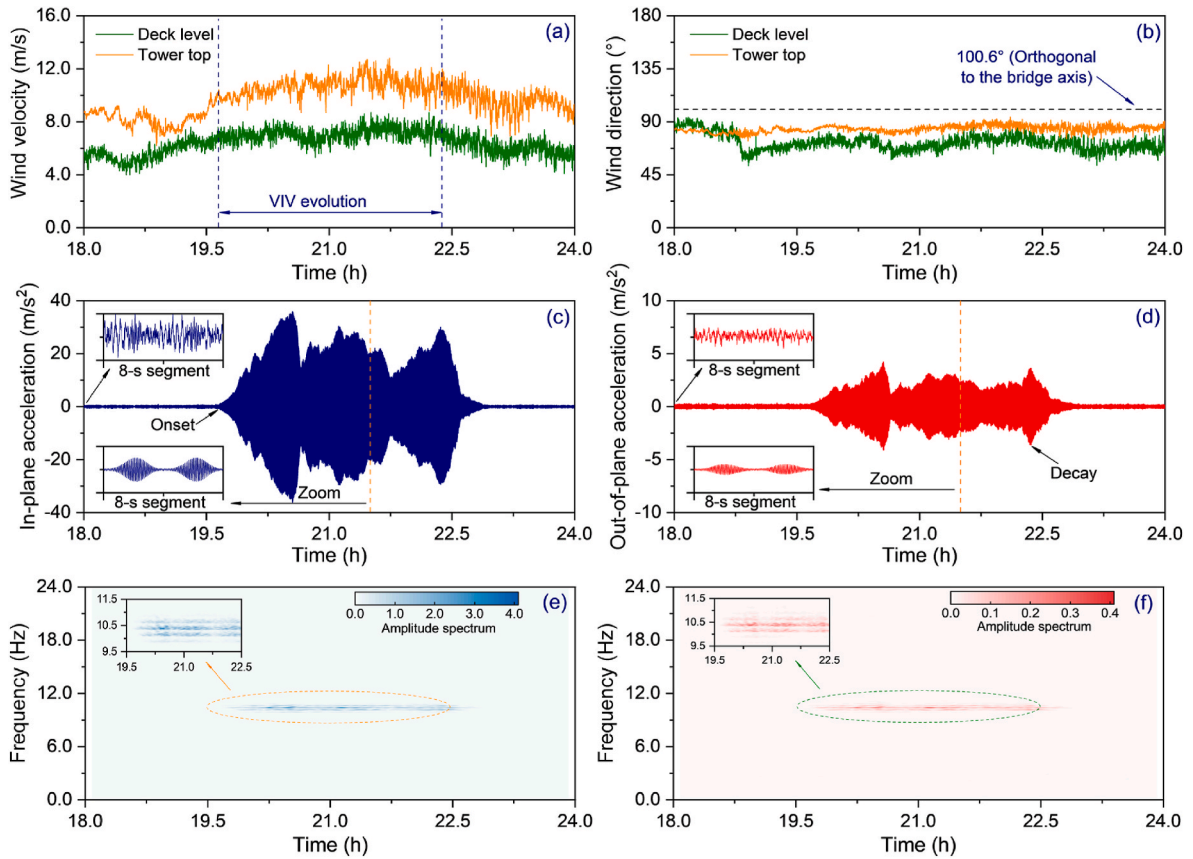


Fig. 7. Vortex-induced vibration event of the stay cable on September 13, 2019: (a) (b) wind velocities and directions; (c) (d) time histories of the accelerations; (e) (f) evolution of the acceleration spectra.

wind velocity; $R_u(\tau)$ is the autocorrelation function of the wind velocity. The analysis is taken with a basic interval of 10 min.

Fig. 9 illustrates the effects of windy conditions on the modal responses of the stay cable, which is obtained through the AMD and modal coordinate amplification (see Eqs. (10)–(12)). The results show that the vibrations predominantly occur in the 41st to 43rd modes, while the

responses in other modes (e.g., the 40th and 44th modes) are considerably smaller. On the other hand, as depicted in Fig. 9a, the large-amplitude vibrations are mainly within a range of wind velocity U_{tower} from 7.0 to 14.0 m/s (or equivalently $U_{deck} = 4.4$ –8.8 m/s). The maximum occurs at the wind velocity $U_{tower} = 10.8$ m/s (or equivalently $U_{deck} = 6.8$ m/s), corresponding to an averaged reduced velocity $U/fD =$

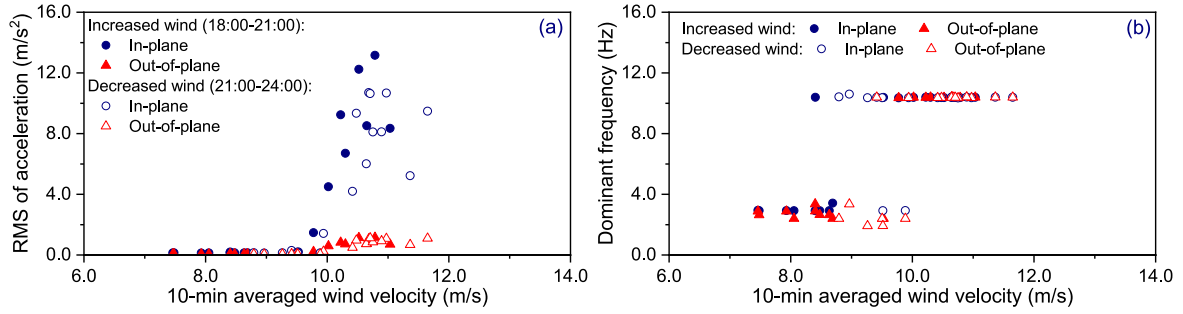


Fig. 8. Relationship between the tower-top wind velocity and the vibration response of the stay cable during the vortex-induced vibration event on September 13, 2019: (a) 10-min RMS (root mean square) of the acceleration; (b) dominant vibration frequency.

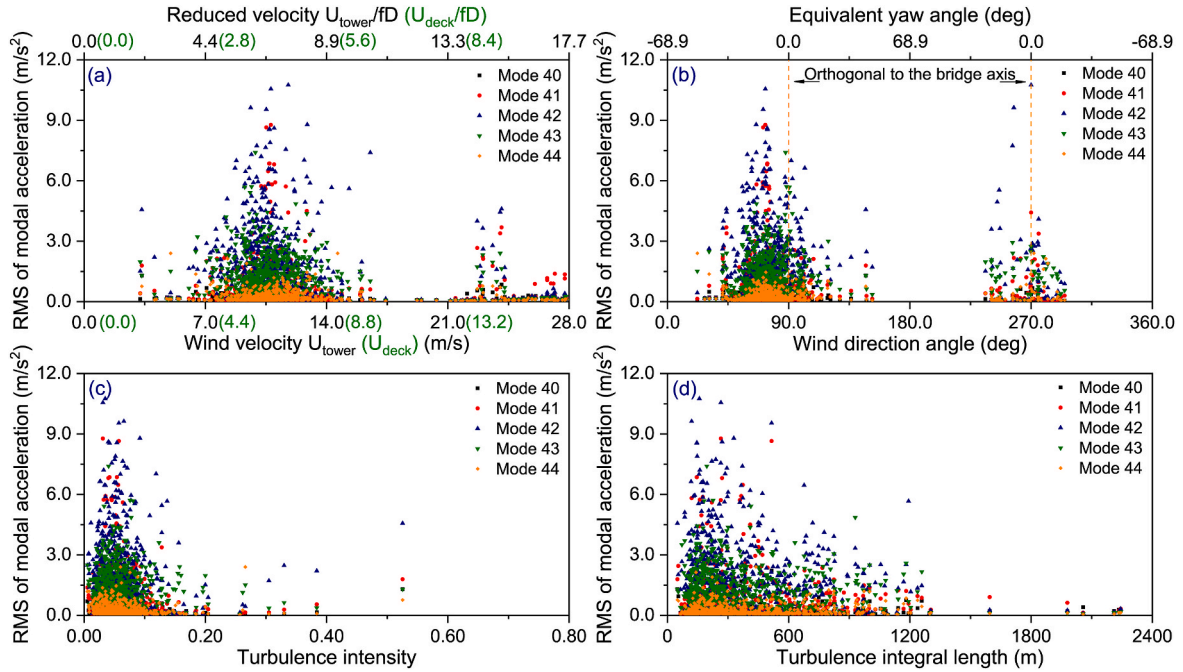


Fig. 9. Effects of the windy conditions on the 10-min RMS of the in-plane modal accelerations of the stay cable during the vortex-induced vibrations observed in 2019: (a) wind velocity; (b) relative wind direction; (c) turbulence intensity; (d) turbulence integral length. Note that, in (a), the deck-level wind velocities presented are determined based on the gradient wind coefficient referring to the tower-top wind velocities; the reduced velocity is determined by referring to $f = 10.4$ Hz, i.e., the natural frequency of the 42nd mode.

5.6 with $f = 10.4$ Hz, i.e., the natural frequency of the 42nd mode. As seen in Fig. 9b, the cable vibrations are prone to occur under the winds perpendicular to the bridge axis (or the stay cable plane), but they are not restricted solely to the orthogonal winds. Specifically, the VIV can be observed in two wide regions of the relative wind direction angle, i.e., 45° – 115° and 240° – 290° . This suggests that there still are noticeable vibration responses even at the wind yaw angle reaching 45° , which is similar to the experimental observations by Liu et al. (2021). Fig. 9c and d demonstrate the dependence of the cable vibrations on the incoming flow turbulence. The vibration events are found mostly linked with relatively smooth flows with turbulence intensity below 0.10. More significant vibrations occur at the turbulence intensity of approximately 0.05. Additionally, the vibrations can be observed over a wide range of turbulence integral lengths from 50 m to 1200 m. These lengths are considerably greater than the diameter of the stay cable, which could potentially reduce the sensitivity of the vibrations to the turbulence integral length.

3.3. Results of damping identification

The above analysis focuses on the response characteristics of the stay cable during the VIVs. Whereas, the associated damping features deserve further investigation. The mechanical damping of the stay cable is first identified for determining the aerodynamic damping subsequently. To this end, the ambient-excited responses of the stay cable are selected at enough small wind velocities, i.e., $U_{tower} < 4.0$ m/s, to decrease the wind effects as much as possible (Li et al., 2011). To reduce the errors due to possible uncertainties, six sets of 4-h acceleration signals, recorded on different days in 2019, are herein employed. Fig. 10 displays the time histories and PSD functions of both in-plane and out-of-plane accelerations in a case. On this basis, the present method is utilized to identify the multi-mode mechanical damping ratios of the stay cable. The first fifty single-mode free-decay responses are determined by applying the AMD and RDT successively. In the RDT process, the trigger level is set as 1.5 times the standard deviation of the decomposed acceleration responses, as suggested by Wen et al. (2018), and the duration of the signal segment is about 25 cycles of the decays. Subsequently, the mechanical damping parameters are identified by

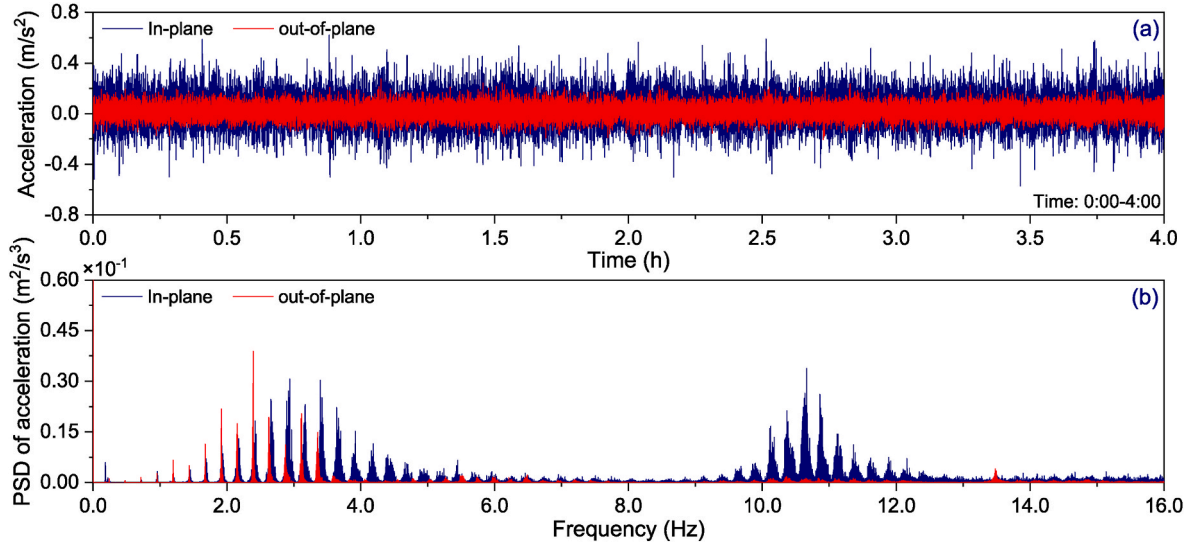


Fig. 10. Ambient-excited accelerations of the stay cable on April 25, 2019: (a) time histories; (b) power spectrum densities.

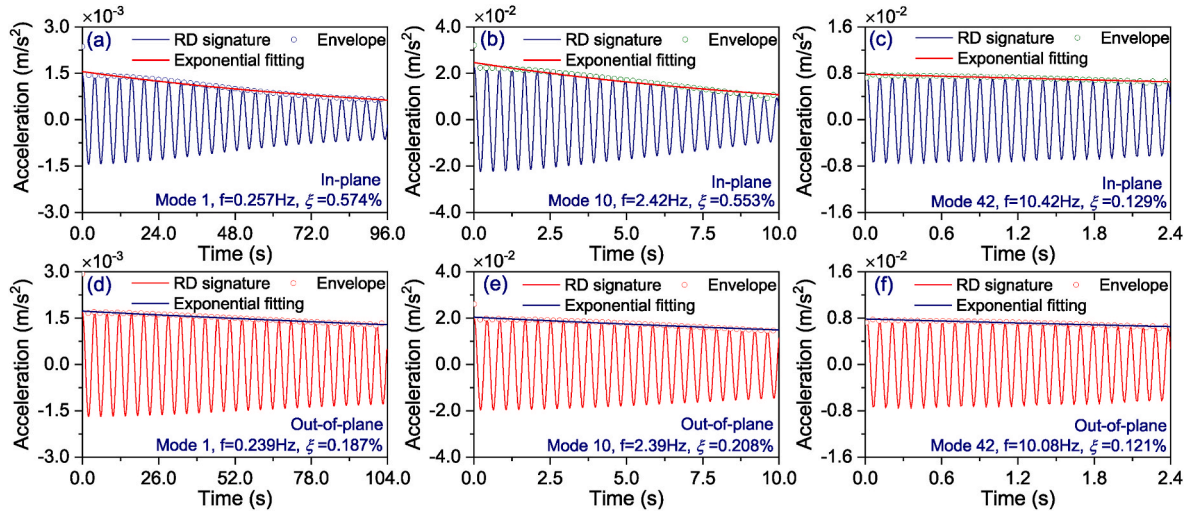


Fig. 11. Free decay responses and their envelopes of the three individual modes identified based on the measured acceleration of the stay cable on April 25, 2019.

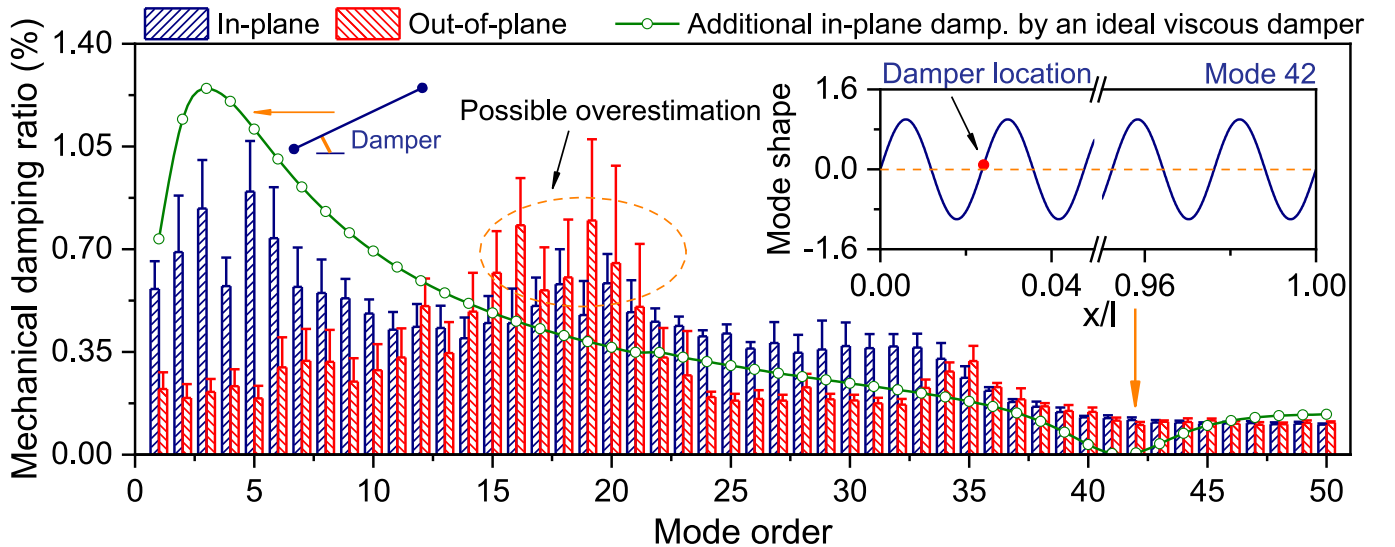


Fig. 12. Estimated mechanical damping ratios of the stay cable based on the measured response under ambient excitations.

combining the HT with the linear least-square fitting, as shown in Fig. 11. The satisfactory fits achieved illustrate that the stay cable can be approximately considered as a linear mechanical system at small amplitudes.

Accordingly, the identified mechanical damping ratios of the stay cable for the first fifty modes are summarized in Fig. 12, including the means and standard deviations. The in-plane and out-of-plane fundamental-mode mechanical damping ratios are estimated to be 0.56% and 0.22%, respectively. The in-plane mechanical damping ratio is significantly larger than the out-of-plane one, which is primarily ascribed to the MR damper equipped in the cable plane. The out-of-plane fundamental-mode mechanical damping ratio (without the MR damper's effect) is close to, but slightly greater than, the previously-reported results for other stay cables in the in-plane fundamental modes, e.g., 0.178% in Ni et al. (2007) and approximately 0.16% in Yamaguchi and Fujino (1998), confirming the reliability of the identified results. Furthermore, the in-plane mechanical damping ratio is generally decreased in higher modes. For the high modes above the 40th order, both in-plane and out-of-plane mechanical damping ratios are only approximately 0.1%. These mechanical dampings should be primarily attributed to the inherent damping properties of the stay cable itself and the MR damper as well as the high-damping rubber. Moreover, to characterize the effects of the external damper, the additional in-plane mechanical damping ratios by an ideal viscous damper optimized targeting the third mode, determined via the iterative scheme (Krenk, 2000), are also drawn in Fig. 12. It is observed that, beyond the optimal mode, the additional mechanical damping ratio gradually decreases as the mode order increases, which is consistent with the field-identified results. In addition, by comparing with the theoretical results, it is noticed that the mechanical damping ratios around the 18th mode may be overestimated due to the weak signals in these modes (see Fig. 10), while those in the 45th to 50th modes may be underestimated as the MR damper does not effectively detect the high-mode ambient-excited vibration with small

displacements.

Subsequently, the total damping of the cable during the VIVs can be identified using the proposed method, as shown in Fig. 13. Based on the observed typical VIV event (see Fig. 7), multiple groups of individual-mode responses are firstly extracted from the overall signal. The attached spectra demonstrate that the single-mode signal is successfully obtained after using the AMD. This decomposition also shows that the vibrations manifest initially on the 42nd and 43rd modes, 12 min earlier than that of the 41st mode. In addition, the decay of the vibrations varies among the three modes, and the 42nd mode endures over 10 min longer than the other two modes. Therefore, the 42nd mode is believed to play a more dominant role in this VIV event. Moreover, based on the smoothing spline fittings of the modal displacement envelopes, the time-varying total damping ratios are determined in this figure. It is found that the maximum negative total damping ratios occur around the onset instants, which are identified as 0.095‰, 0.11‰, and 0.098‰ for the 41st, 42nd, and 43rd in-plane modes, respectively. As the vibration amplitudes increase, the negative total damping ratios generally decrease. In the fully-developed stage, the total damping ratios are relatively small with slight fluctuations. During the decaying stage, the positive total dampings increase as the vibration amplitudes decrease, and the maximum positive total dampings are usually accompanied by the end of the decays. Overall, the total damping ratios are relatively small, which results in the VIV development being so slow that it spans approximately 1 h. Moreover, the in-plane and out-of-plane motions have almost the same total dampings, indicating that the two motions share similar evolution patterns. More specifically, the out-of-plane motion follows the in-plane motion.

Based on the mechanical and total damping identified above, aerodynamic damping of the stay cable in the VIVs can be obtained to illustrate the in-situ aeroelasticity. Fig. 14 shows the aerodynamic damping against the modal displacement amplitude during the growth-to-resonance stage of the vibration. Interestingly, several closed loops

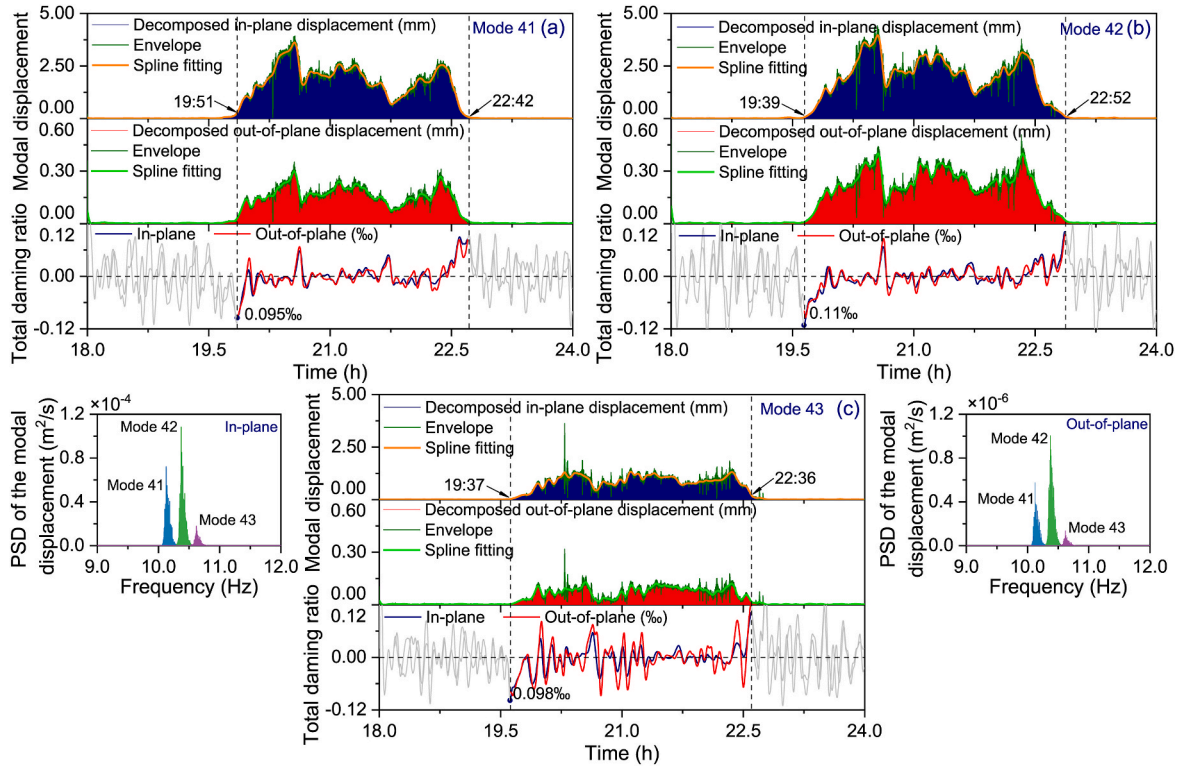


Fig. 13. Identification of the time-varying total damping ratios of the stay cable during the vortex-induced vibration event on September 13, 2019: (a) the 41st mode; (b) the 42nd mode; (c) the 43rd mode. Note that the measured accelerations have been numerically integrated to yield the displacements via the frequency-domain algorithm (Liu et al., 2022) before the identification.

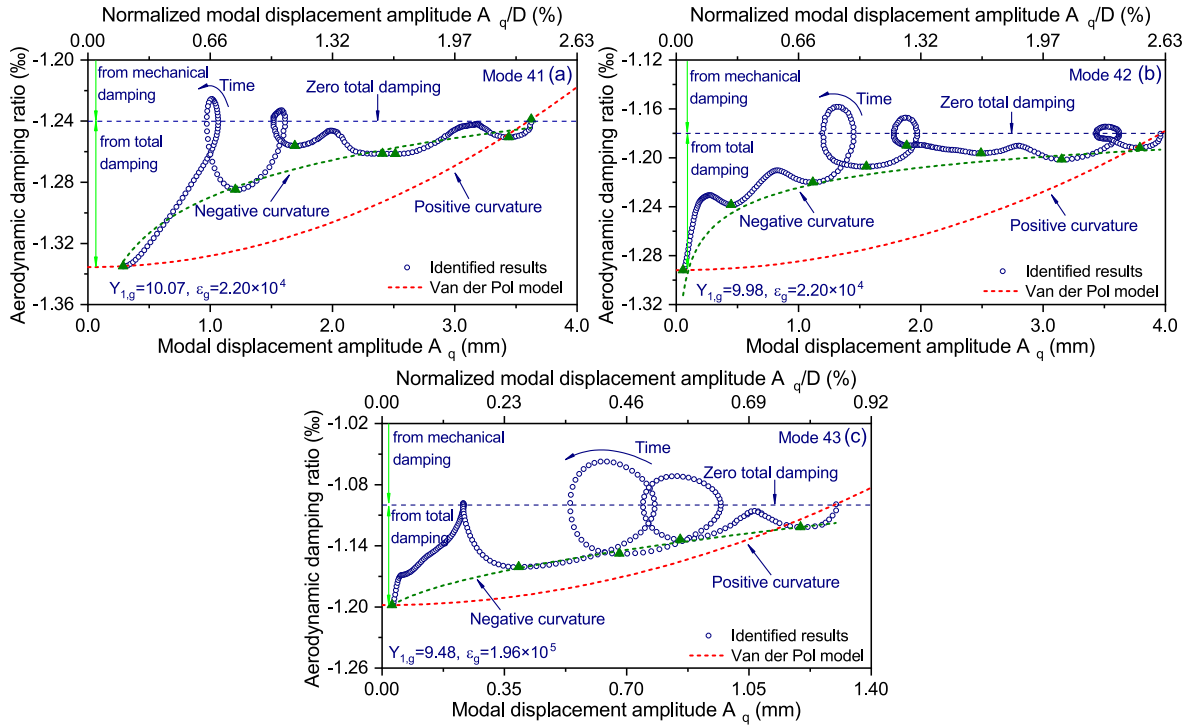


Fig. 14. Identified amplitude-dependent aerodynamic damping ratios during the growth-to-resonance stage of the vortex-induced vibration event on September 13, 2019: (a) the 41st mode; (b) the 42nd mode; (c) the 43rd mode. Note that in this figure, “curvature” is defined according to the aerodynamic damping (negative value) rather than the aerodynamic parameter (positive value), and thus its sign is opposite to these in Lupi et al. (2018), Guo et al. (2021) and M. Zhang et al. (2021).

are observed in these curves, resulting in multiple aerodynamic damping values for a given modal displacement amplitude. The direct cause for this phenomenon is that the vibration evolution may experience local downs in growth rate (i.e., total damping) and even decrease but then continue to increase progressively (see Fig. 13). Therefore, the aerodynamic damping, or more specifically total damping, may exhibit ups and downs over time, as denoted by the arrays in this figure. The inherent reason for this phenomenon is that the aerodynamic damping is not solely determined by the vibration amplitude, but varies with the windy condition, etc. Specifically, slight variations in wind velocity and/or direction may significantly alter the vibration evolution. In addition, multi-mode coupling may influence the surrounding flow fields and thereby fluctuate aerodynamic damping. Overall, the aerodynamic damping of an in-situ flexible cable is more complicated than that of a sectional cable model from experimental/numerical tests.

More importantly, if ignoring these loops, the aerodynamic damping can be roughly characterized by their local minima denoted in dots, which represent stronger exciting capability around this amplitude to some extent. In this sense, the negative aerodynamic damping is found to generally decrease as the displacement amplitude increases. This decrease is mainly due to the nonlinear aeroelastic effects that the vortex-induced forces are significantly dependent on the oscillation amplitudes (Lupi et al., 2018; R. Zhang et al., 2023). Moreover, the van der Pol-type polynomials are applied to model the variation of the aerodynamic damping ratio against the modal displacement amplitude, according to the principle of capturing the maximum aerodynamic damping at the initial small-amplitude vibration and the amplitude at zero total damping, as shown in Fig. 14. Accordingly, the linear and nonlinear generalized aerodynamic parameters are obtained as follows: $Y_{1,g} = 10.07, 9.98$, and 9.48 and $\varepsilon_g = 2.20 \times 10^4, 2.20 \times 10^4$ and 1.96×10^5 for the 41st to 43rd modes with $K_{ref} = 1.16, 1.19$ and 1.22 , respectively. Also, it is noticed that, in the van der Pol model, the negative aerodynamic damping decays faster at a larger vibration amplitude, characterized as a positive curvature ($d^2\varepsilon_a/dA_q^2 > 0$), whereas, the field-identified aerodynamic damping seems to decay more

slowly, i.e., a negative curvature ($d^2\varepsilon_a/dA_q^2 < 0$). Similar findings regarding the curvature are also seen in the experimental and numerical investigations of circular cylinders (Lupi et al., 2018; Guo et al., 2021; M. Zhang et al., 2021; R. Zhang et al., 2023). Therefore, it is necessary to develop more advanced models for nonlinear aerodynamic damping in VIVs (Larsen, 1995; M. Zhang et al., 2020, 2021; Guo et al., 2021), but this aspect is not the focus herein.

3.4. Discussion

With the identified damping parameters, the VIVs of the stay cable can be re-analyzed, aiming to reveal the mechanism for the onset of the vibrations and to give a control strategy.

3.4.1. On onset mechanism

In general, wind-induced cable vibrations could occur if the negative aerodynamic damping, related to Karman vortices or wind-rain action, exceeds the corresponding mechanical damping. For VIVs, the Karman vortices could be locked mode-by-mode as the wind velocity increases, indicating potential excitations in all concerned modes. However, in the present observations, the VIVs only or primarily exhibit in the 41st to 43rd modes, rather than mode-by-mode excitations. The reasons are mainly two-fold: (i) The vortex-induced negative aerodynamic damping is relatively small, having a very weak capability to overcome the mechanical damping of the stay cable. For example, the maximum negative aerodynamic damping during the typical VIV event is only 1.3%, approximately. (ii) The stay cable exhibits considerable mechanical damping, derived from the MR damper, the high-damping rubber and the cable itself, in most modes, even including the high-order modes (e. g., above the 45th mode), as demonstrated in the measured and analytical results in Fig. 12. However, several particular high-order modes are exceptions, due to these modes exhibiting a shape node close to the MR damper (see Fig. 12). Consequently, the additional mechanical damping by the damper is almost zero for these modes. Therefore, the VIVs are observed only within the particular frequency

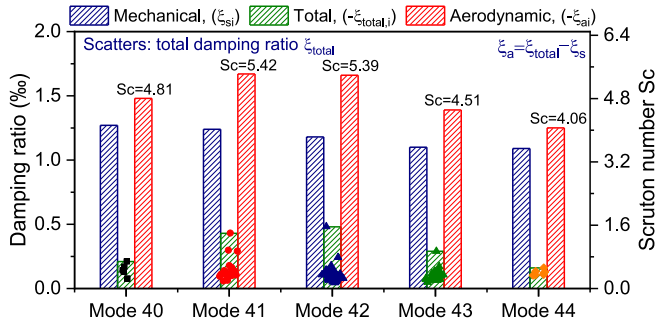


Fig. 15. Suggested Scruton number thresholds for mitigating the high-order multi-mode vortex-induced vibrations based on the identified damping parameters of the stay cable.

bands where the damper is located on the mode shape nodes. It is also expected that the VIVs could be observed in a wider frequency band without the MR damper.

In addition, it is interesting to discuss a phenomenon in practice that VIVs are prone to occur in long cables, whereas rarely observed in relatively short cables. An important factor is the shorter cables have a larger fundamental frequency, which results in the onset velocity for these “particular” modes being above the wind velocity encountered commonly at the bridge sites.

3.4.2. On control criterion

More importantly, the identified aerodynamic damping can be in turn used to determine the mechanical damping required for mitigating VIVs. To improve the generality, the maximum negative aerodynamic damping ratios are identified from the recorded VIVs in 2019, as summarized in Fig. 15. The scatters are primarily in the 41st to 43rd modes, with less in the 40th and 44th modes, because the vibrations are predominantly triggered in the 41st to 43rd modes. Moreover, for convenience in engineering practice, the extreme aerodynamic damping is further equivalently normalized to be the Scruton number (Sc) that unifies the aerodynamic damping ratio and the mass ratio:

$$Sc > \frac{m|\xi_a|}{\rho D^2} \quad (23)$$

Accordingly, the obtained Sc threshold is shown in Fig. 15, characterizing the extreme capability of the vortex-induced loads to excite the stay cable into vibrations. The Sc thresholds for the 40th to 44th modes are 4.81, 5.42, 5.39, 4.51, and 4.06, respectively. If the Sc number of the stay cable itself ($Sc = m\xi_s/\rho D^2$) exceeds these thresholds, VIVs will not occur. Overall, based on the damping identifications, this study suggests a Scruton number criterion of $Sc > 5.4$ for mitigating the VIVs of stay cables, which can be further applied used for the optimal design of damper systems (Di et al., 2021; Wang et al., 2022b). Also, note that the current Sc threshold, determined by the maximum negative aerodynamic damping at the onset of the VIVs, is capable of controlling VIVs completely, rather than reducing the VIV amplitude to a certain level. Partly due to this, the Sc threshold is larger than that derived from the aerodynamic parameters of sectional cylinders at normalized amplitudes of 0.03 ($Sc = 1.33$ (Lupi et al., 2018),) and 0.05 ($Sc = 1.16$ (R. Zhang et al., 2023),).

On the other hand, it is necessary to indicate that the criterion of $Sc > 10$ is widely adopted for controlling wind-induced cable vibration, particularly for RWIV (Irwin, 1997), which is much larger than the present results. It implies that the Karman vortices, compared with the rain-wind action, are weaker in exciting cables into vibrations. In other words, mitigating VIV needs a smaller mechanical damping. In addition, the present criterion for VIVs is different from that for dry-cable galloping where the Sc threshold increases with the critical velocity required (Jafari and Sarkar, 2019, 2020).

4. Conclusions

This study proposed an analytical method to identify the nonlinear aerodynamic damping associated with multi-mode wind-induced cable vibrations in field environments. For this method, a series of individual-mode transient wind-induced resonances and free-to-decay signatures are firstly extracted from the monitored signals through the AMD and RDT, and then the total and mechanical damping, as well as the corresponding aerodynamic damping, can be estimated using the HT combined with the smoothing spline and linear least-square fitting. Moreover, based on the one-year field monitoring on the Sutong Bridge, the method was applied to address the response characteristics, onset mechanism, and control strategies of the multi-mode VIVs of an ultra-long stay cable ($L = 551.01\text{m}$, $D = 0.152\text{m}$, $L/D = 3625$) with an external damper.

The monitoring results showed that the stay cable experienced high-frequency vibrations, mainly involving the 41st to 43rd modes (around 10.4 Hz). The vibrations were featured by the non-planar motions, with dominant in-plane motion and accompanied out-of-plane motion. The vibration mechanism was detected as VIV according to the Strouhal law. The statistical analysis revealed that the VIVs mainly occurred within the wind velocity range of $U_{\text{tower}} = 7.0\text{--}14.0\text{ m/s}$ or equivalently $U_{\text{deck}} = 4.4\text{--}8.8\text{ m/s}$ and across wide wind direction ranges (e.g., the yaw angle reaching 45°). The VIVs were mostly associated with relatively smooth flows with turbulence intensity below 0.10 but were not very sensitive to the turbulence integral length. Subsequently, by applying the proposed method, the mechanical and time-varying total damping ratios were identified based on the measured ambient-excited responses and the resonance evolution, respectively, yielding the corresponding aerodynamic damping ratios. The results showed that the high-mode mechanical damping ratios are notably small, i.e., approximately 1.0‰ for the modes above the 40th order. The aerodynamic damping exhibits a maximum negative value at the onset instant and then decreases generally with increasing oscillation amplitude. The amplitude-dependent aerodynamic damping was roughly described by the van der Pol model with nonlinear aerodynamic parameters. Finally, the mechanism for the onset was revealed, attributing to the weak mechanical damping for the specific modes with a shape node encountered by the external damper. More importantly, based on the identified extreme value of aerodynamic damping, a criterion of $Sc > 5.4$ was proposed for mitigating the VIVs of stay cables, which is helpful for the optimal design of damper systems.

This study only focused on applying the analytical method to investigate the multi-mode VIVs of a stay cable. In fact, the method can be extended to analyze other types of vibrations, such as RWIV and galloping of stay cables, as well as VIVs on the decks of cable-stayed and suspension bridges (Xu et al., 2019). In this extension, the mode orders of concern should be lower than those in the present case, and better results may be achieved due to larger frequency intervals.

CRedit authorship contribution statement

Ruilin Zhang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Zhiwen Liu:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Conceptualization. **Shuai Zhou:** Writing – review & editing, Formal analysis. **Yafei Wang:** Investigation, Formal analysis. **Zhengqing Chen:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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